Infinite Sequences and Series

9.1 Sequences and Series

A **sequence** is an ordered list of numbers

$$a_1,\ a_2,\ a_3,\ \ldots,\ a_n,\ \ldots$$

whereas a **series** is an infinite sum of a list of numbers

$$a_1 + a_2 + a_3 + \cdots + a_n + \cdots.$$

Mathematically, a sequence is defined as a function whose domain is the set of positive integers.

If a sequence $\{a_n\}$ has the **limit** L, where L is a real number, it is written as

$$\lim_{n\to\infty} a_n = L,$$

and we say the sequence **converges** to L. If the sequence does not have a limit we say the sequence **diverges**.

Given a sequence of numbers of $\{a_n\}$, an expression of the form

$$\sum_{n=1}^{\infty} a_n = a_1 + a_2 + a_3 + \cdots + a_n + \cdots$$

is an **infinite series**.

For the infinite series $\sum a_n$, the ***n*th partial sum** is given by

$$s_n = \sum_{i=1}^{n} a_i = a_1 + a_2 + a_3 + \cdots + a_n.$$

If the sequence of partial sums converges to a limit L, we say that the series **converges**, and its sum is L. If the series does not converge, we say that the series **diverges**.

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The **geometric series** $a + ar + ar^2 + \cdots + ar^{n-1} + \cdots$ converges to the sum

$$\sum_{n=1}^{\infty} ar^{n-1} = \frac{a}{1-r}, \text{ if } |r| < 1.$$

If $|r| \geq 1$, the series diverges.

A series is a **telescoping series** if it is in the form

$$(a_1 - a_2) + (a_2 - a_3) + (a_3 - a_4) + (a_4 - a_5) + \cdots.$$

Since the *n*th partial sum of the series is $S_n = a_1 - a_{n+1}$, a telescoping series converges if and only if $\lim_{n\to\infty} a_n = L$, where L is a real number.

The sum of the series is $S = a_1 - L$.

Limit of *n*th Term of a Convergent Series

If the series $\sum_{n=1}^{\infty} a_n$ converges, then $\lim_{n\to\infty} a_n = 0$.

The converse of this theorem is not true in general. But the contrapositive of this theorem provides a useful test for divergent series.

***n*th Term Test for Divergence**

If $\lim_{n\to\infty} a_n$ does not exist or $\lim_{n\to\infty} a_n \neq 0$, then the series $\sum_{n=1}^{\infty} a_n$ diverges.

Exercises

geometric series

1. $\sum_{n=1}^{\infty} \frac{(3)^{n+1}}{5^n} =$

(A) $\frac{3}{5}$ (B) $\frac{5}{2}$ (C) $\frac{9}{2}$ (D) The series diverges

2. If $f(x) = \sum_{n=1}^{\infty} (\tan x)^n$, then $f(1) =$

(A) −2.794 (B) −0.61 (C) 0.177 (D) The series diverges

3. $\sum_{n=2}^{\infty} \frac{2}{n^2 - 1} =$

(A) 0 (B) $\frac{1}{2}$ (C) 1 (D) $\frac{3}{2}$

4. The sum of the geometric series $\frac{2}{21} + \frac{4}{63} + \frac{8}{189} + \ldots$ is

(A) $\frac{5}{21}$ (B) $\frac{2}{7}$ (C) $\frac{4}{7}$ (D) The series diverges

5. If $S_n = \left(\frac{3^{n-1}}{(4+n)^{20}}\right)\left(\frac{(7+n)^{20}}{3^n}\right)$, to what number does the sequence $\{S_n\}$ converge?

(A) $\frac{1}{3}$ (B) $\frac{7}{4}$ (C) $\left(\frac{7}{4}\right)^{20}$ (D) Diverges

Free Response Questions

8. Find the sum of the series $\sum_{n=1}^{\infty} \left(\frac{3}{n(n+3)} + \frac{1}{7^n}\right)$

Telescop

geometric

9.2 The Integral Test and *p*-Series

The Integral Test

If f is positive, continuous, and decreasing on $[1,\infty)$ and $a_n = f(n)$, then

$$\sum_{n=1}^{\infty} a_n \quad \text{and} \quad \int_1^{\infty} f(x)\,dx$$

either both converge or both diverge. In other words:

1. If $\sum_{n=1}^{\infty} a_n$ is convergent, then $\int_1^{\infty} f(x)\,dx$ is convergent.
2. If $\sum_{n=1}^{\infty} a_n$ is divergent, then $\int_1^{\infty} f(x)\,dx$ is divergent.

***p*- Series and Harmonic Series**

The *p*- series $\sum_{n=1}^{\infty} \frac{1}{n^p} = \frac{1}{1^p} + \frac{1}{2^p} + \frac{1}{3^p} + \frac{1}{4^p} + \cdots$

is convergent if $p > 1$ and divergent if $0 < p \le 1$.

For $p = 1$, the series $\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots$ is called **harmonic series**.

A **general harmonic series** is of the form $\sum \frac{1}{(an+b)}$.

Exercises

1. If $\int_1^{\infty} \frac{dx}{x^2+1} = \frac{\pi}{4}$, then which of the following must be true?

I. $\sum_{n=1}^{\infty} \frac{1}{n^2+1}$ diverges.

II. $\sum_{n=1}^{\infty} \frac{1}{n^2+1}$ converges.

III. $\sum_{n=1}^{\infty} \frac{1}{n^2+1} = \frac{\pi}{4}$

(A) none (B) I only (C) II only (D) II and III only

2. Which of the following series converge?

I. $\sum_{n=1}^{\infty} \frac{n}{2n^2+1}$ II. $\sum_{n=1}^{\infty} ne^{-n^2}$ III. $\sum_{n=2}^{\infty} \frac{1}{x \ln x}$

(A) I only (B) II only (C) III only (D) I and II only

3. What are all values of p for which $\sum_{n=1}^{\infty} \frac{\sqrt{n}}{n^p+1}$ converges?

(A) $p>0$ (B) $p>\frac{1}{2}$ (C) $p>1$ (D) $p>\frac{3}{2}$

4. What are all values of k for which the series $1+(\sqrt{2})^k+(\sqrt{3})^k+(\sqrt{4})^k+\cdots+(\sqrt{n})^k+\cdots$ converges?

(A) $k<-2$ (B) $k<-1$ (C) $k>1$ (D) $k>2$

5. Determine whether the series is convergent or divergent.

(a) $\sum_{n=1}^{\infty} \frac{1}{n^2+1}$ (b) $\sum_{n=1}^{\infty} \frac{\ln n}{n}$

6. Determine whether the series is convergent or divergent.

(a) $1+\frac{1}{\sqrt[3]{4}}+\frac{1}{\sqrt[3]{9}}+\frac{1}{\sqrt[3]{16}}+\frac{1}{\sqrt[3]{25}}+\cdots$

(b) $\sum_{n=1}^{\infty} n^{1-\pi}$

9.3 The Comparison Test

Direct Comparison Test

Let $0 < a_n \le b_n$ for all n.

1. If $\sum_{n=1}^{\infty} b_n$ converges, then $\sum_{n=1}^{\infty} a_n$ converges.
2. If $\sum_{n=1}^{\infty} a_n$ diverges, then $\sum_{n=1}^{\infty} b_n$ diverges.

Limit Comparison Test

If $a_n > 0$, $b_n > 0$, and $\lim_{n\to\infty} \frac{a_n}{b_n} = L$, where L is finite and positive, then both series either converge or both diverge.

Note: When choosing a series for comparison, you can disregard all but the highest powers of n in both the numerator and denominator.

Exercises

1. Determine whether the series is convergent or divergent.

(a) $\sum_{n=2}^{\infty} \frac{n}{n^2 - 3}$

(b) $\sum_{n=1}^{\infty} \frac{\sin^2 n}{\sqrt{n^3 + 1}}$

2. Determine whether the series is convergent or divergent.

(a) $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n^2 + 4}}$

(b) $\sum_{n=3}^{\infty} \frac{2^n}{3^n + 1}$

3. Which of the following series diverge?

I. $\sum_{n=1}^{\infty} \frac{1}{n!}$

II. $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n} + 2}$

III. $\sum_{n=1}^{\infty} \sin(\frac{1}{n})$

(A) I only (B) II only (C) II and III only (D) I, II, and III

4. Which of the following series converge?

I. $\sum_{n=1}^{\infty} \frac{n^{3/2}}{3n^3 + 7}$

II. $\sum_{n=1}^{\infty} \frac{1}{\sqrt[3]{n^4 + 1}}$

III. $\sum_{n=1}^{\infty} \frac{n!}{n^n}$

(A) I only (B) I and II only (C) I and III only (D) I, II, and III

5. Which of the following series cannot be shown to converge using the limit comparison test with the series $\sum_{n=1}^{\infty} \frac{1}{2^n}$?

(A) $\sum_{n=1}^{\infty} \frac{1}{2^n - 1}$

(B) $\sum_{n=1}^{\infty} \frac{n}{2^n}$

(C) $\sum_{n=1}^{\infty} \frac{2n}{2^{n+1}\sqrt{n^2+1}}$

(D) $\sum_{n=1}^{\infty} \frac{2n^2 - 3n}{2^n(n^2 + n - 100)}$

6. Determine whether the following series converge or diverge.

(a) $\sum_{n=1}^{\infty} \frac{\cos(2n)}{1 + (1.6)^n}$

(b) $\sum_{n=1}^{\infty} \frac{4^n}{2^n + 3^n}$

9.4 Alternating Series and Error Bound

Alternating Series Test

Let $a_n > 0$. The alternating series

$$\sum_{n=1}^{\infty} (-1)^n a_n \quad \text{and} \quad \sum_{n=1}^{\infty} (-1)^{n+1} a_n$$

converge if the following two conditions are met.

1. $\lim_{n \to \infty} a_n = 0$ 2. $a_{n+1} \le a_n$, for all n greater than some integer N.

Alternating Series Estimation Theorem (Error Bound)

If S_n is a partial sum and $S = \sum_{n=1}^{\infty} (-1)^n a_n$ is the sum of a convergent alternating series that satisfies the condition $a_{n+1} \le a_n$, then the remainder $R_n = S - S_n$ is smaller than a_{n+1}, which is the absolute value of the first neglected term.

$$|R_n| = |S - S_n| \le a_{n+1}$$

Definition of Absolute and Conditional Convergence

1. $\sum a_n$ is **absolutely convergent** if $\sum |a_n|$ converges.
2. $\sum a_n$ is **conditionally convergent** if $\sum a_n$ converge but $\sum |a_n|$ diverges.

Exercises

1. Determine whether the series is convergent or divergent.

(a) $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}}$ (b) $\sum_{n=1}^{\infty} (-1)^n \frac{n}{2n-1}$

2. Determine whether the series is absolutely convergent, conditionally convergent, or divergent.

(a) $\sum_{n=1}^{\infty} \frac{(-1)^n \sqrt[n]{e}}{n^2}$ (b) $\sum_{n=1}^{\infty} (-1)^{n+1} n^{-2/3}$

3. Let $f(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots + - \frac{(-1)^n x^{2n}}{(2n)!} + \cdots$.

Use the alternating series error bound to show that $1 - \frac{1}{2!} + \frac{1}{4!}$ approximates $f(1)$ with an error less than $\frac{1}{500}$.

$$f(1) = 1 - \frac{1}{2!} + \frac{1}{4!} - \frac{1}{6!} + \cdots + \frac{(-1)^n}{(2n)!} + \cdots$$

Since series is alternating, with terms convergent to 0 and decreasing in absolute value, the error is less than the first neglected term.

So, $\left| f(1) - (1 - \frac{1}{2!} + \frac{1}{4!}) \right| \le \frac{1}{6!} = \frac{1}{720} < \frac{1}{500}$.

4. Which of the following series converge?

I. $\sum_{n=1}^{\infty} \frac{(-1)^n \sqrt{n}}{n}$ II. $\sum_{n=1}^{\infty} \frac{(-1)^n}{\ln n}$ III. $\sum_{n=1}^{\infty} \cos(n\pi)$

(A) I only (B) II only (C) III only (D) I and II only

5. Which of the following series converge?

I. $\sum_{n=1}^{\infty} (-1)^n \cos(\frac{\pi}{n})$ II. $\sum_{n=1}^{\infty} \sin(\frac{2n-1}{2})\pi$ III. $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{2n}{n^2+1}$

(A) I only (B) II only (C) III only (D) I and II only

6. For what integer k, $k > 1$, will both $\sum_{n=1}^{\infty} \frac{(-1)^{kn}}{\sqrt{n}}$ and $\sum_{n=1}^{\infty} \frac{n^2\sqrt{n}}{n^k+1}$ converge?

(A) 3 (B) 4 (C) 5 (D) 6

7. Let $s=\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n^3}$ and s_n be the sum of the first n terms of the series. If $|s-s_n|<\frac{1}{500}$ what is the smallest value of n?

(A) 6 (B) 7 (C) 8 (D) 9

8. Which of the following statements about the series $\sum_{n=1}^{\infty}\frac{(-1)^{n+1}(n+3)}{n^2}$ is true?

(A) The series converges conditionally.

(B) The series converges absolutely.

(C) The series converges but neither conditionally nor absolutely.

(D) The series diverges.

9. Let $f(x)=1-\frac{3x}{2!}+\frac{9x^2}{4!}-\frac{27x^3}{6!}+\cdots+\frac{(-1)^n(3x)^n}{(2n)!}+\cdots$.

Use the alternating series error bound to show that $1-\frac{3}{2!}+\frac{9}{4!}$ approximates $f(1)$ with an error less than $\frac{1}{20}$.

9.5 The Ratio Test

Ratio Test

Let $\sum a_n$ be a series with nonzero terms.

1. $\sum a_n$ converges absolutely if $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| < 1$.

2. $\sum a_n$ diverges if $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| > 1$ or $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \infty$.

3. The Ratio Test is inconclusive if $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = 1$.

Exercises

1. Determine whether the series is convergent or divergent.

(a) $\sum_{n=1}^{\infty}\frac{3^n}{n!}$ (b) $\sum_{n=1}^{\infty}(-1)^n\frac{n^3}{5^n}$ (c) $\sum_{n=1}^{\infty}\frac{3^n}{2^n-1}$

2. Determine whether the series is conditionally convergent or absolute convergent.

(a) $\sum_{n=1}^{\infty}(-1)^n\frac{\sqrt{n}}{n+3}$ (b) $\sum_{n=1}^{\infty}\frac{(-1)^n e^n}{n!}$

3. Which of the following series converge?

I. $\sum_{n=1}^{\infty}\frac{n!}{2^n}$ II. $\sum_{n=1}^{\infty}\frac{n}{3^n}$ III. $\sum_{n=1}^{\infty}n\left(\frac{2}{3}\right)^n$

(A) I only (B) II only (C) II and III only (D) I, II, and III

4. Which of the following series converge?

I. $\sum_{n=1}^{\infty}\frac{n!}{n^n}$ II. $\sum_{n=1}^{\infty}\frac{(n!)^2}{(2n)!}$ III. $\sum_{n=1}^{\infty}\frac{n^9}{9^n}$

(A) I only (B) II only (C) I and II only (D) I, II, and III

5. Determine whether the following series converge or diverge.

(a) $\sum_{n=1}^{\infty} \frac{n!}{n\,2^n}$

(b) $\sum_{n=0}^{\infty} \frac{\cos^n x}{2^n}$

(c) $\sum_{k=1}^{\infty} \frac{3^k\,k!}{(k+3)!}$

Summary of Tests for Series

Test	Series	Conditions of Convergence or Divergence
n th-Term	$\sum_{n=1}^{\infty} a_n$	The series is divergent if $\lim_{x\to\infty} a_n \neq 0$. Test is inconclusive if $\lim_{x\to\infty} a_n = 0$.
Geometric Series	$\sum_{n=1}^{\infty} ar^{n-1}$	The series is convergent if $\lvert r\rvert<1$, divergent if $\lvert r\rvert \geq 1$. $S = \frac{a}{1-r}$
Telescoping Series	$\sum_{n=1}^{\infty} (a_n - a_{n+1})$	The series is convergent if $\lim_{x\to\infty} a_n = L$. $S = a_1 - L$
p-Series	$\sum_{n=1}^{\infty} \frac{1}{n^p}$	The series is convergent if $p > 1$, divergent if $p \leq 1$.
Alternating Series	$\sum_{n=1}^{\infty} (-1)^{n-1} a_n$	The series is convergent if $\lim_{x\to\infty} a_n = 0$ and $0 < a_{n+1} \leq a_n$.
Integral	$\sum_{n=1}^{\infty} a_n$	If f is positive, continuous, and decreasing for $x \geq 1$ and $a_n = f(n)$, then the series converges if $\int_1^{\infty} f(x)\,dx$ converges, diverges if $\int_1^{\infty} f(x)\,dx$ diverges.
Ratio	$\sum_{n=1}^{\infty} a_n$	The series is convergent if $\lim_{n\to\infty} \left\lvert \frac{a_{n+1}}{a_n} \right\rvert < 1$, divergent if $\lim_{n\to\infty} \left\lvert \frac{a_{n+1}}{a_n} \right\rvert > 1$, inconclusive if $\lim_{n\to\infty} \left\lvert \frac{a_{n+1}}{a_n} \right\rvert = 1$.
Direct Comparison	$\sum_{n=1}^{\infty} a_n$	Let $0 < a_n \leq b_n$ for all n. If $\sum_{n=1}^{\infty} b_n$ converges, then $\sum_{n=1}^{\infty} a_n$ converges. If $\sum_{n=1}^{\infty} a_n$ diverges, then $\sum_{n=1}^{\infty} b_n$ diverges.
Limit Comparison	$\sum_{n=1}^{\infty} a_n$	Suppose that $a_n > 0$, $b_n > 0$, and $\lim_{n\to\infty} (\frac{a_n}{b_n}) = L$. Then $\sum_{n=1}^{\infty} a_n$ converges if $\sum_{n=1}^{\infty} b_n$ converges and $\sum_{n=1}^{\infty} a_n$ diverges if $\sum_{n=1}^{\infty} b_n$ diverges.

Power Series

A **power series about** $x = 0$ is an infinite series of the form

$$\sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots + a_n x^n + \cdots$$

where x is a variable and the a_n's are constants.

More generally, a series of the form

$$\sum_{n=0}^{\infty} a_n (x-c)^n = a_0 + a_1 (x-c) + a_2 (x-c)^2 + \cdots + a_n (x-c)^n + \cdots$$

is called a **power series centered at** c, or a **power series about** c.

Convergence of a Power Series

For a power series centered at c there are only three possibilities:

1. The series converges only at $x = c$.
2. The series converges for all x.
3. There exists a real number $R > 0$ such that the series converges for $|x-c| < R$, and diverges for $|x-c| > R$.

The number R is called the **radius of convergence** of the power series.

In most cases, R can be found by using the Ratio Test. The Ratio Test always fails when x is an endpoint of the interval of convergence, so each endpoint must be tested separately for convergence or divergence.

If the series converges only at c, the radius of convergence is $R = 0$.
If the series converges for all x, the radius of convergence is $R = \infty$.
The set of all values of x for which the power series converges is the **interval of convergence** of the power series.

Exercises

1. Find the radius of convergence and interval of convergence of the series $\sum_{n=0}^{\infty} \frac{(-2)^n x^n}{\sqrt{n+3}}$.

2. Find the radius of convergence and interval of convergence of the series $\sum_{n=0}^{\infty} n!(2x)^n$.

3. What are all values of x for which the series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^n}{n^3}$ converges?

(A) $-1 < x < 1$ (B) $-1 \le x \le 1$ (C) $-1 < x \le 1$ (D) $-1 \le x < 1$

4. What are all values of x for which the series $\sum_{n=0}^{\infty} \frac{x^{n+1}}{(n+1)!}$ converges?

(A) $0 < x < 2$ (B) $0 \le x < 2$ (C) $-1 < x \le 2$ (D) All real x

5. What are all values of x for which the series $\sum_{n=2}^{\infty} \frac{(-1)^n x^n}{2^n \sqrt{n}}$ converges?

(A) $-2 < x < 2$ (B) $-2 \le x < 2$ (C) $-2 < x \le 2$ (D) All real x

6. What are all values of x for which the series $\sum_{n=1}^{\infty} n!(3x-2)^n$ converges?

(A) No values of x (B) $(-\infty, \frac{2}{3}]$ (C) $x = \frac{2}{3}$ (D) $[\frac{2}{3}, \infty)$

7. Find the radius of convergence and the interval of convergence for the series

$$\sum_{n=1}^{\infty} \frac{2 \cdot 4 \cdot 6 \cdots (2n)}{1 \cdot 3 \cdot 5 \cdots (2n-1)} x^{n+1}.$$

Geometric Power Series

A power series representation for $f(x)=\dfrac{1}{1-x}$ can be obtained from the sum of a geometric series $\sum_{n=0}^{\infty} ar^n = \dfrac{a}{1-r}$, if you let $a=1$ and $r=x$.

Therefore $\dfrac{1}{1-x}=\sum_{n=0}^{\infty} x^n = 1+x+x^2+x^3+\cdots, \quad |x|<1$.

Example 1 □ Find the first four nonzero terms and the general terms for the power series expansion of the function given by $f(x)=\dfrac{2}{1+x^2}$.

Solution □ $f(x)=\dfrac{2}{1+x^2}=\dfrac{2}{1-(-x^2)}$ is a geometric series with $a=2$ and $r=-x^2$.

Therefore, $f(x)=2-2x^2+2x^4-2x^6+\cdots+(-1)^n 2x^{2n}+\cdots$.

Differentiation and Integration of Power Series

If the function given by

$$f(x)=\sum_{n=0}^{\infty} a_n(x-c)^n$$
$$=a_0+a_1(x-c)+a_2(x-c)^2+\cdots+a_n(x-c)^n+\cdots$$

is differentiable, the derivative and antiderivative of f are as follows.

1. $$f'(x)=\sum_{n=0}^{\infty} na_n(x-c)^{n-1}$$
$$=a_1+2a_2(x-c)+3a_3(x-c)^2+\cdots$$

2. $$\int f(x)=C+\sum_{n=0}^{\infty} a_n\frac{(x-c)^{n+1}}{n+1}$$
$$=C+a_0(x-c)+a_1\frac{(x-c)^2}{2}+a_2\frac{(x-c)^3}{3}+\cdots$$

The radius of convergence of the series obtained by differentiating or integrating a power series is the same as that of the original power series. However the interval of convergence may differ as a result of the behavior at the endpoints.

Exercises

1. Let f be a function given by $f(x) = \sum_{n=1}^{\infty} \frac{(-1)^n (x-2)^n}{n}$.

Find the interval of convergence for each of the following.

(a) $f(x)$ (b) $f'(x)$ (c) $\int f(x)\,dx$.

2. The power series expansion for $\frac{1}{1-x}$ is $\sum_{n=0}^{\infty} x^n$. Which of the following is a power series expansion for $\frac{1}{1+x^3}$?

(A) $1+x^2+x^4+x^6+\cdots$

(B) $1-x^3+x^6-x^9+\cdots$

(C) $1+\frac{x^3}{3}+\frac{x^6}{6}+\frac{x^9}{9}+\cdots$

(D) $1-\frac{x^3}{3}+\frac{x^6}{6}-\frac{x^9}{9}+\cdots$

3. The power series expansion for $\frac{1}{1-x}$ is $\sum_{n=0}^{\infty} x^n$. Which of the following is a power series expansion for $\frac{1}{2-x}$?

(A) $1+\frac{x}{2}+\frac{x^2}{4}+\frac{x^3}{8}+\cdots$

(B) $1-\frac{x}{2}+\frac{x^2}{4}-\frac{x^3}{8}+\cdots$

(C) $\frac{1}{2}+\frac{x}{4}+\frac{x^2}{8}+\frac{x^3}{16}+\cdots$

(D) $\frac{1}{2}-\frac{x}{4}+\frac{x^2}{8}-\frac{x^3}{16}+\cdots$

4. If $f(x) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{(x-2)^n}{n!} = (x-2) - \frac{(x-2)^2}{2!} + \frac{(x-2)^3}{3!} - \frac{(x-2)^4}{4!} + \cdots$, which of the following represents $f'(x)$?

(A) $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{(x-2)^{n-1}}{n!}$

(B) $\sum_{n=1}^{\infty} (-1)^{n} \frac{(x-2)^{n-1}}{(n+1)!}$

(C) $\sum_{n=0}^{\infty} (-1)^{n+1} \frac{(x-2)^{n-1}}{n!}$

(D) $\sum_{n=0}^{\infty} (-1)^{n} \frac{(x-2)^{n}}{n!}$

5. A power series expansion for $f(x) = \frac{1}{1-x}$ can be obtained from the sum of the geometric series $\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r}$, if you let $a = 1$ and $r = x$.. Let $g(x)$ be defined as $g(x) = \frac{1}{1+x}$.

(a) Write the first four terms and the general term of the power series expansion of $g(x)$.

(b) Write the first four terms and the general term of the power series expansion of $g(x^2)$.

(c) Write the first four terms and the general term of the power series expansion of h, where $h(x) = \int g(x^2)\, dx$ and $h(0) = 0$.

(d) Find the value of $h(1)$.

nth Taylor Polynomial and nth Maclaurin Polynomial

If a function f has n derivatives at c, then the polynomial

$$P_n(x) = f(c) + f'(c)(x-c) + \frac{f''(c)}{2!}(x-c)^2 + \frac{f'''(c)}{3!}(x-c)^3 + \cdots + \frac{f^{(n)}(c)}{n!}(x-c)^n$$

is called the **nth Taylor polynomial for f at c**. If $c = 0$, then

$$P_n(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \cdots + \frac{f^{(n)}(0)}{n!}x^n$$

is called the **Maclaurin polynomial for f**.

Guidelines for Finding a Taylor Polynomial

1. Differentiate $f(x)$ several times and evaluate each derivative at c.

   $f(c),\ f'(c),\ f''(c),\ f'''(c),\ \cdots,\ f^{(n)}(c)$

2. Use the sequence developed in the first step to form the **Taylor coefficients**

   $a_n = \dfrac{f^{(n)}(c)}{n!}$.

Example 1 □ Let f be the function given by $f(x) = \ln(2-x)$. Write the third-degree Taylor polynomial for f about $x = 1$ and use it to approximate $f(1.2)$.

Solution □

$f(x) = \ln(2-x)$ $\qquad$ $f(1) = \ln(2-1) = \ln 1 = 0$

$f'(x) = \dfrac{-1}{2-x}$ $\qquad$ $f'(1) = \dfrac{-1}{2-1} = -1$

$f''(x) = \dfrac{-1}{(2-x)^2}$ $\qquad$ $f''(1) = \dfrac{-1}{(2-1)^2} = -1$

$f'''(x) = \dfrac{-2}{(2-x)^3}$ $\qquad$ $f'''(1) = \dfrac{-2}{(2-1)^2} = -2$

$$P_3(x) = f(1) + f'(1)(x-1) + \frac{f''(1)}{2!}(x-1)^2 + \frac{f'''(1)}{3!}(x-1)^3$$

$$= -(x-1) - \frac{1}{2}(x-1)^2 - \frac{1}{3}(x-1)^3$$

$$f(1.2) \approx P_3(1.2) = -(1.2-1) - \frac{1}{2}(1.2-1)^2 - \frac{1}{3}(1.2-1)^3$$

$$\approx -.222666$$

Exercises

1. Let $P(x)=\frac{1}{3}-\frac{2}{3}x+\frac{2}{3}x^2-\frac{4}{9}x^3+\frac{2}{9}x^4$ be the fourth-degree Taylor polynomial for the function f about $x=0$. What is the value of $f^{(4)}(0)$?

(A) $-\frac{32}{3}$ (B) $-\frac{4}{3}$ (C) $\frac{8}{9}$ (D) $\frac{16}{3}$

2. Let $P(x)=4-3x^2+\frac{13}{12}x^4-\frac{121}{360}x^6$ be the sixth-degree Taylor polynomial for the function f about $x=0$. What is the value of $f'''(0)$?

(A) $-\frac{121}{15}$ (B) $-\frac{3}{2}$ (C) 0 (D) $\frac{121}{15}$

3. Let f be a function that has derivatives of all orders for all real numbers. If $f(1)=2$, $f'(1)=-3$, $f''(1)=4$, and $f'''(1)=-9$, which of the following is the third-degree Taylor polynomial for f about $x=1$?

(A) $P(x)=2-3(x-1)+2(x-1)^2-\frac{3}{2}(x-1)^3$

(B) $P(x)=2-3(x+1)+2(x+1)^2-\frac{3}{2}(x+1)^3$

(C) $P(x)=2-3(x-1)+4(x-1)^2-9(x-1)^3$

(D) $P(x)=2-3(x+1)+2(x+1)^2-3(x+1)^3$

4. The third-degree Taylor polynomial of xe^x about $x=0$ is

(A) $P_3(x)=x-\frac{1}{2}x^2+\frac{1}{6}x^3$

(B) $P_3(x)=x+x^2+\frac{1}{2}x^3$

(C) $P_3(x)=x+x^2-\frac{1}{3}x^3$

(D) $P_3(x)=1-x+\frac{1}{2}x^2-\frac{1}{6}x^3$

5. The second-degree Taylor polynomial of $\sec x$ about $x=\frac{\pi}{4}$ is

(A) $P_2(x)=1+\sqrt{2}(x-\frac{\pi}{4})+\sqrt{2}(x-\frac{\pi}{4})^2$

(B) $P_2(x)=\frac{\sqrt{2}}{2}+\frac{\sqrt{2}}{2}(x-\frac{\pi}{4})+\frac{3\sqrt{2}}{3!}(x-\frac{\pi}{4})^2$

(C) $P_2(x)=\sqrt{2}+\sqrt{2}(x-\frac{\pi}{4})+\frac{3\sqrt{2}}{2!}(x-\frac{\pi}{4})^2$

(D) $P_2(x)=1+\sqrt{2}(x-\frac{\pi}{4})+\frac{3\sqrt{2}}{3!}(x-\frac{\pi}{4})^2$

6. Let $P(x)=3-2(x-2)+5(x-2)^2-12(x-2)^3+3(x-2)^4$ be the fourth-degree Taylor polynomial for the function f about $x=2$. Assume f has derivatives of all orders for all real numbers.

(a) Find $f(2)$ and $f^{\prime\prime\prime}(2)$.

(b) Write the third-degree Taylor polynomial for f' about 2 and use it to approximate $f'(2.1)$.

(c) Write the fourth-degree Taylor polynomial for $g(x)=\int_2^x f(t)\,dt$ about 2.

(d) Can $f(1)$ be determined from the information given? Justify your answer.

Taylor Series and Maclaurin Series

If a function f has derivatives of all orders at $x = c$, then the series

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!}(x-c)^n$$

$$= f(c) + f'(c)(x-c) + \frac{f''(c)}{2!}(x-c)^2 + \frac{f'''(c)}{3!}(x-c)^3 + \cdots + \frac{f^{(n)}(c)}{n!}(x-c)^n + \cdots$$

is called the **Taylor series for $f(x)$ at c**. Moreover, if $c = 0$, then the series is called the **Maclaurin series for f**.

Example 1 □ Find the Maclaurin series for the function $f(x) = \ln(1+x)$.

Solution □

$f(x) = \ln(1+x)$ $\qquad f(0) = 0$

$f'(x) = \dfrac{1}{1+x}$ $\qquad f'(0) = 1$

$f''(x) = -\dfrac{1}{(1+x)^2}$ $\qquad f''(0) = -1$

$f'''(x) = \dfrac{2}{(1+x)^3}$ $\qquad f'''(0) = 2$

$f^{(4)}(x) = -\dfrac{6}{(1+x)^4}$ $\qquad f^{(4)}(0) = -6$

$$\ln(1+x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \cdots + \frac{f^{(n)}(0)}{n!}x^n + \cdots$$

$$= 0 + 1 \cdot x + \frac{-1}{2!}x^2 + \frac{2}{3!}x^3 + \frac{-6}{4!}x^4 + \cdots + \frac{(-1)^{n-1}(n-1)!}{n!}x^n + \cdots$$

$$= x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \frac{1}{4}x^4 + \cdots + \frac{(-1)^{n-1}}{n}x^n + \cdots$$

Power Series for Elementary Functions

Function	Interval of Convergence
$\frac{1}{x} = 1-(x-1)+(x-1)^2-(x-1)^3+\cdots+(-1)^n(x-1)^n+\cdots$	$0<x<2$
$\frac{1}{1-x} = 1+x+x^2+x^3+\cdots+x^n+\cdots$	$-1<x<1$
$\ln x = (x-1)-\frac{(x-1)^2}{2}+\frac{(x-1)^3}{3}-\cdots+\frac{(-1)^{n-1}(x-1)^n}{n}+\cdots$	$0<x\le 2$
$e^x = 1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\cdots+\frac{x^n}{n!}+\cdots$	$-\infty<x<\infty$
$\sin x = x-\frac{x^3}{3!}+\frac{x^5}{5!}-\frac{x^7}{7!}+\cdots+\frac{(-1)^n x^{2n+1}}{(2n+1)!}+\cdots$	$-\infty<x<\infty$
$\cos x = 1-\frac{x^2}{2!}+\frac{x^4}{4!}-\frac{x^6}{6!}+\cdots+\frac{(-1)^n x^{2n}}{(2n)!}+\cdots$	$-\infty<x<\infty$
$\tan^{-1} x = x-\frac{x^3}{3}+\frac{x^5}{5}-\frac{x^7}{7}+\cdots+\frac{(-1)^n x^{2n+1}}{2n+1}+\cdots$	$-1\le x\le 1$

Multiplication of Power Series

Power series can be multiplied the way we multiply polynomials. We usually find only the first few terms because the calculations for the later terms become tedious and the initial terms are the most important ones.

Exercises

1. Find the Maclaurin series for the function $f(x) = \cos x^2$.

2. Find the Maclaurin series for the function $f(x) = x^2 e^x - x^2$.

3. A series expansion of $\frac{\arctan x}{x}$ is

(A) $1-\frac{x}{3}+\frac{x^3}{5}-\frac{x^5}{7}+\cdots$

(B) $1-\frac{x^2}{3}+\frac{x^4}{5}-\frac{x^6}{7}+\cdots$

(C) $1-\frac{x^2}{2!}+\frac{x^4}{4!}-\frac{x^6}{6!}+\cdots$

(D) $x-\frac{x^3}{3}+\frac{x^4}{5}-\frac{x^6}{7}+\cdots$

4. The coefficient of x^3 in the Taylor series for e^{-2x} about $x=0$ is

(A) $-\frac{4}{3}$ (B) $-\frac{2}{3}$ (C) $-\frac{1}{3}$ (D) $\frac{4}{3}$

5. A function f has a Maclaurin series given by $-\frac{x^4}{3!}+\frac{x^6}{5!}-\frac{x^8}{7!}+\cdots+\frac{(-1)^n x^{2n+2}}{(2n+1)!}+\cdots$.

Which of the following is an expression for $f(x)$?

(A) $x^3e^x-x^2$

(B) $x\ln x-x^2$

(C) $\tan^{-1}x-x$

(D) $x\sin x-x^2$

6. A series expansion of $\frac{x-\sin x}{x^2}$ is

(A) $\frac{1}{2!}-\frac{x^2}{4!}+\frac{x^4}{6!}+\cdots+\frac{(-1)^{n+1}x^{2n-2}}{(2n)!}+\cdots$

(B) $\frac{x}{2!}-\frac{x^3}{4!}+\frac{x^5}{6!}+\cdots+\frac{(-1)^{n+1}x^{2n+1}}{(2n)!}+\cdots$

(C) $\frac{x}{3!}-\frac{x^3}{5!}+\frac{x^5}{7!}+\cdots+\frac{(-1)^{n+1}x^{2n-1}}{(2n+1)!}+\cdots$

(D) $\frac{x^2}{3!}-\frac{x^4}{5!}+\frac{x^6}{7!}+\cdots+\frac{(-1)^{n+1}x^{2n}}{(2n+1)!}+\cdots$

7. $\sum_{n=0}^{\infty}\frac{(-1)^n x^{n+2}}{n!}$ is the Taylor series about zero for which of the following functions?

(A) $x\sin x$ (B) $x\cos x$ (C) x^2e^{-x} (D) $x\ln(x+1)$

8. Let f be the function given by $f(x)=e^{-x}$.

(a) Write the first four terms and the general term of the Taylor series for f about $x=0$.

(b) Use the result from part (a) to write the first four nonzero terms and the general term of the series expansion about $x=0$ for $g(x)=\frac{1-x-f(x)}{x}$.

(c) For the function g in part (b), find $g'(-1)$ and use it to show that $\sum_{n=1}^{\infty}\frac{n}{(n+1)!}=1$.

Coordinate-System Conversion Formula

If a point P has rectangular coordinates (x, y) and polar coordinates (r, θ), then

$$x = r\cos\theta, \qquad y = r\sin\theta,$$

$$r^2 = x^2 + y^2, \qquad \tan\theta = \frac{y}{x}(x \neq 0).$$

The **slope** of the tangent line to the graph of $r = f(\theta)$ at point (r, θ) is

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{\frac{d}{d\theta}(r\sin\theta)}{\frac{d}{d\theta}(r\cos\theta)} = \frac{\frac{dr}{d\theta}\sin\theta + r\cos\theta}{\frac{dr}{d\theta}\cos\theta - r\sin\theta}, \text{ where } \frac{dx}{d\theta} \neq 0 \text{ at } (r, \theta).$$

If $r > 0$ and $\frac{dr}{d\theta} < 0$ for $\alpha < \theta < \beta$, then r is decreasing on this interval, which means the curve is getting closer to the origin.

If $r > 0$ and $\frac{dr}{d\theta} > 0$ for $\alpha < \theta < \beta$, then r is increasing on this interval, which means the curve is getting farther from the origin.

If $\frac{dr}{d\theta} = 0$, the curve is either closest to the origin or farthest from the origin.

Area in Polar Coordinates

The area of the region bounded by the graph of $r = f(\theta)$ between the radial lines $\theta = \alpha$ and $\theta = \beta$ is given by

$$A = \frac{1}{2}\int_{\alpha}^{\beta} [f(\theta)]^2 d\theta$$

$$= \frac{1}{2}\int_{\alpha}^{\beta} r^2 d\theta$$

Exercises

1. If $r = \theta - 3\sin\theta$ then $\frac{dr}{d\theta}$ at (π, π) is

(A) 2 (B) π (C) 4 (D) 2π

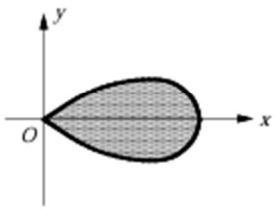
2. If $r = \frac{2}{1-\cos\theta}$ then $\frac{dr}{d\theta}$ at $(2, \frac{\pi}{2})$ is

(A) -2 (B) -1 (C) 0 (D) 1

3. If $r = 3\sin\theta$ then $\dfrac{dy}{dx}$ at the point where $\theta = \dfrac{\pi}{3}$ is

(A) -2 (B) $-\sqrt{3}$ (C) -1 (D) $\dfrac{\sqrt{3}}{3}$

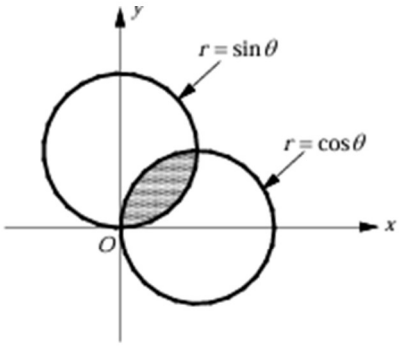
4.



What is the area of the region enclosed by the loop of the graph of the polar curve $r = 2\cos(2\theta)$ shown in the figure above?

(A) $\dfrac{\pi}{4}$ (B) $\dfrac{\pi}{2}$ (C) $\dfrac{3\pi}{4}$ (D) π

5.



The area of the shaded region that lies inside the polar curves $r = \sin\theta$ and $r = \cos\theta$ is

(A) $\dfrac{1}{8}(\pi - 2)$ (B) $\dfrac{1}{4}(\pi - 2)$ (C) $\dfrac{1}{2}(\pi - 2)$ (D) $\dfrac{1}{8}(\pi - 1)$

6. The area of the region enclosed by the polar curve $r = 2 + \sin\theta$ is

(A) 3π (B) $\dfrac{7\pi}{2}$ (C) 4π (D) $\dfrac{9\pi}{2}$

7. Which of the following gives the area of the region inside the polar curve $r = 1 + \cos\theta$ and outside the polar curve $r = 2\cos\theta$?

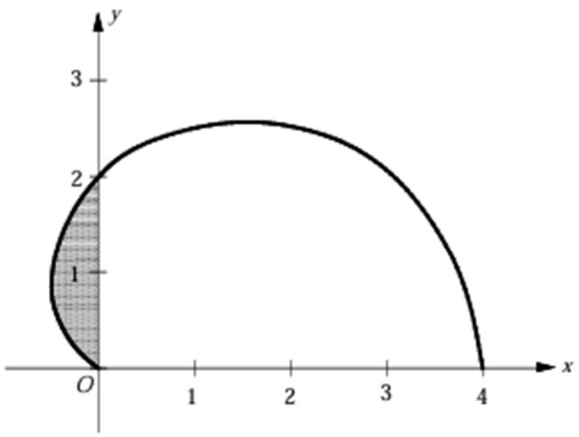
(A) $\dfrac{1}{2}\displaystyle\int_0^{2\pi} (1 - \cos\theta)^2 \, d\theta$

(B) $\displaystyle\int_0^{2\pi} (1 + 2\cos\theta)^2 \, d\theta$

(C) $\displaystyle\int_0^{2\pi} (1 + \cos\theta)^2 \, d\theta - \pi$

(D) $\dfrac{1}{2}\displaystyle\int_0^{2\pi} (1 + \cos\theta)^2 \, d\theta - \pi$

8.



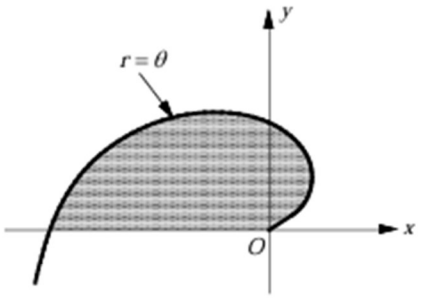
. The graph of the polar curve $r = 2 + 2\cos(\theta)$ for $0 \le \theta \le \pi$ is shown above.

(a) Write an integral expression for the area of the shaded region.

(b) Write expressions for $\frac{dx}{d\theta}$ and $\frac{dy}{d\theta}$ in terms of θ.

(c) Write an equation in terms of x and y for the line tangent to the curve at the point where $\theta = \frac{\pi}{2}$.

9.



The area of the shaded region bounded by the polar curve $r = \theta$ and the x-axis is

(A) $\frac{\pi^2}{4}$ (B) $\frac{\pi^3}{6}$ (C) $\frac{\pi^3}{3}$ (D) $\frac{\pi^3}{2}$